USN | | | | | | | | | | [24MTSE101]

# FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

## STRUCTURAL DYNAMICS

**Time:3 Hours** **Maximum Marks:100**

**Instructions:**
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries20 marks.
ii. Missing data may be suitably assumed.

| | | | CO's | Blooms Level | Marks |
|---|---|---|---|---|---|
| **1.** | **a.** | Derive the equation of motion for damped free vibration and explain the response for under-damped, critically damped, and over-damped systems. | **1** | **2** | **10** |
| | **b.** | Describe the mathematical model of single degree of freedom system. Explain the difference between free and forced vibration in the context of SDOF systems. | **1** | **2** | **10** |
| | | | | | |
| **2.** | **a.** | Explain the Newmark-beta method in detail, including the formulation of the equations and the role of the parameters β and γ in controlling the accuracy and stability of the method | **2** | **3** | **10** |
| | **b.** | Describe Duhamel's integral and its importance in analyzing transient responses of SDOF systems. Compare the central difference method and Newmark-beta method in terms of stability and accuracy. | **2** | **3** | **10** |
| | | | | | |
| **3.** | **a.** | A two-story building can be modelled as a 2-DOF system with the following parameters: m1=300 kg, m2=200 kg, k1=50,000 N/m, k2=30,000 N/m. Find the natural frequencies and mode shapes of the system | **3** | **3** | **10** |
| | **b.** | Explain the concept of degrees of freedom in the context of multi-degree-of-freedom (MDOF) systems. Briefly explain how natural frequencies and mode shapes are determined for a 2-DOF system. | **3** | **3** | **10** |
| | | **OR** | | | |
| **4.** | **a.** | Derive the equations of motion for an MDOF system using the mass and stiffness matrix method. Explain how eigenvalue analysis is used to find natural frequencies and mode shapes. | **3** | **3** | **10** |
| | **b.** | What is the orthogonality property of mode shapes? Why is it important in modal analysis? What are the assumptions made in the shear building model for multi-story structures? | **3** | **3** | **10** |
| | | | | | |
| **5.** | **a.** | Derive Rayleigh's quotient and use it to estimate the fundamental natural frequency of a simply supported beam with given mass and stiffness properties | **4** | **4** | **10** |
| | **b.** | What are the advantages of using Dunkerley's method in vibration analysis? Define the mass matrix and stiffness matrix in the context | **4** | **4** | **10** |

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| | | of discretized beam vibration | | | |
| | | **OR** | | | |
| **6.** | **a.** | Estimate the fundamental natural frequency of a simply supported beam of length 5m with modulus of elasticity $E = 200 \times 10^9$ N/m$^2$ , moment of inertia $I = 2 \times 10^{-6}$ m$^4$ and mass per unit length $\mu = 50$ kg/m | **4** | **4** | **10** |
| | **b.** | Briefly explain Rayleigh's method for estimating natural frequencies of continuous systems. Explain how lumped and consistent mass matrices differ in finite element analysis. | **4** | **4** | **10** |
| | | | | | |
| **7.** | **a.** | Discuss how the response spectrum method is used to estimate the maximum displacement and acceleration of a building subjected to an earthquake. | **5** | **5** | **10** |
| | **b.** | What is a response spectrum, and how is it used in seismic analysis? How do you compute the seismic response of a structure using the normal mode approach? | **5** | **5** | **10** |
| | | **OR** | | | |
| **8.** | **a.** | Derive the equation of motion for a base-isolated structure subjected to seismic loading. Discuss how base isolation reduces the seismic response. | **5** | **5** | **10** |
| | **b.** | Explain the concept of base isolation in earthquake-resistant design. What is the importance of damping in seismic response analysis? | **5** | **5** | **10** |